<INTRODUCTION TO DATA MANAGEMENT >

PROJECT REPORT

(Project Semester January-April 2025)

(Smart Accident Analytics – A Data-Driven View of Road Safety Trends)

Submitted by

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Programme and Section –B TECH CSE ,K23GW

Course Code - INT217

Under the Guidance of-

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CERTIFICATE

This is to certify that grainy. (student’s name) bearing Registration no. 12316766 has completed INT217 <Course Code> project titled, **“.baljinder kaur ”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor
Designation of the Supervisor
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DECLARATION

I grainy student of .BTECH CSE (Program name) under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date:

Signature

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Acknowledgement

I extend my heartfelt thanks to the BALJINDER KAUR MAM. With data "Smart Accident Analytics – A Data-Driven View of Road Safety Trends" dataset has been an essential resource for my project. I deeply appreciate the hard work and dedication that went into compiling and making this data accessible, as it has allowed me to explore and road safety trends effectively. My gratitude goes out to everyone involved in this effort, which has significantly contributed to the success of my work.

Grainy

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Computer Science and Engineering

Introduction

- In an era where data drives innovation across all sectors, transportation safety is no exception. **Smart Accident Analytics** represents a transformative approach to understanding and improving road safety. By leveraging advanced data analytics, artificial intelligence, and real-time monitoring, this methodology provides deep insights into traffic patterns, accident hotspots, driver behaviour, and environmental conditions that contribute to road incidents.
- Traditionally, accident analysis relied heavily on manual reporting and historical records, often lacking the granularity and timeliness needed for proactive safety interventions. Today, with the proliferation of IoT devices, connected vehicles, and smart city infrastructure, vast amounts of data are being generated every second. **Smart Accident Analytics** harnesses this data to identify trends, predict high-risk scenarios, and support evidence-based policymaking.
- This data-driven view not only helps in reducing the frequency and severity of road accidents but also empowers urban planners, traffic authorities, and policymakers to design safer transportation systems. From dynamic traffic management to automated emergency response systems, Smart Accident Analytics is reshaping the future of road safety—one dataset at a time.
- Road safety remains a critical global concern, with traffic accidents causing significant human, economic, and social losses every year. Despite advancements in vehicle design and traffic regulations, road accidents continue to pose a major challenge, particularly in densely populated and fast-developing urban areas. To address this persistent issue, governments, policymakers, urban planners, and transportation authorities are turning toward innovative solutions that go beyond traditional safety measures. One such emerging approach is Smart Accident Analytics—a data-driven methodology that leverages modern technology to monitor, analyze, and predict road safety trends.
- Smart Accident Analytics involves the collection and analysis of large volumes of traffic-related data from a wide range of sources, including traffic cameras, GPS and telematics systems, vehicle sensors, emergency service records, and social media feeds. These data are processed using advanced analytics techniques such as machine learning, artificial intelligence, geospatial analysis, and statistical modelling to uncover hidden patterns and correlations that can inform proactive safety measures. Rather than relying solely on historical accident data and reactive responses, this approach enables stakeholders to anticipate and mitigate risks before they result in serious incidents.
- With the help of smart analytics, it becomes possible to identify accident-prone locations, determine contributing factors such as road conditions or driver behaviour, and optimize the deployment of traffic enforcement and emergency response resources. For example, predictive models can forecast high-risk periods for traffic collisions based on weather, time of day, and traffic volume. Heat maps and dashboards can visually represent accident clusters, supporting better urban planning and infrastructure development. Additionally, real-time data analysis can assist in improving the responsiveness of emergency services and reducing the time between an accident and medical intervention.

Objectives

The key objectives of this project are as follows:

- Driver & non – motorist substance abuse case
- Weather & surface conditions vs. crash count
- Crash severity by time of day
- Casualties by vehicle types
- Casualties by location

Scope

- This report aims to explore how Smart Accident Analytics is transforming the landscape of road safety through the use of data-driven technologies. It covers the integration of various technologies such as IoT sensors, machine learning, cloud computing, and geospatial analysis in monitoring and analysing road traffic incidents.
- The scope includes an overview of the key data sources involved—ranging from traffic cameras and GPS systems to vehicle telematics and emergency response logs—and how these are combined to generate meaningful insights. The report investigates how analytical techniques are applied to identify accident-prone areas, assess driving behaviours, and predict potential risks, ultimately supporting better policy-making and urban planning.
- In addition to technological foundations, the report discusses practical applications in real-world scenarios, highlighting how cities and transportation agencies use smart analytics to reduce accident rates and improve response times. It also addresses current challenges such as data privacy, system integration, and the need for standardization, while suggesting future directions, including real-time monitoring, AI-powered safety systems, and integration with smart city infrastructure.
- This report provides a well-rounded view of the potential and limitations of Smart Accident Analytics, emphasizing its role in building safer, smarter, and more responsive road networks.

Source of dataset

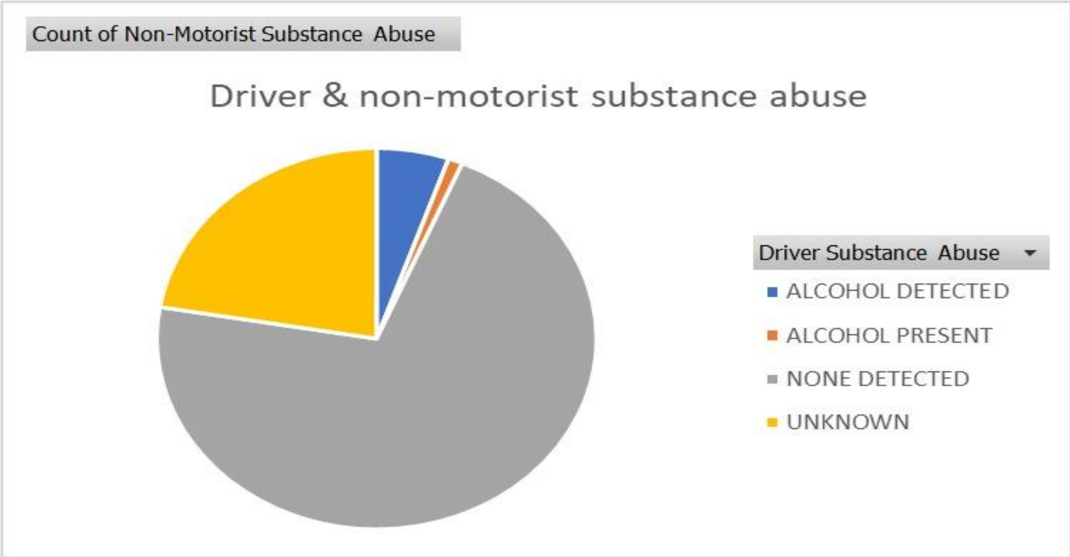
This data is from the data.gov
https://data.montgomerycountymd.gov/api/views/mmzv-x632/rows.csv?accessType=DOWNLOAD

linkdin link

https://www.linkedin.com/posts/grainy-bajaj-7428ba294_exceldashboard-roadsafetyanalytics-datavisualization-activity-7317202285925658625-83y8?utm_source=share&utm_medium=member_desktop&rcm=ACoAAEdTIqIBrGlrdOgIaSugH3lAxx3qcm9dzDs

Analysis on dataset

1. Driver & non-motorist abuse case



- The pie chart titled **“**Driver & Non-Motorist Substance Abuse**”** provides insight into the role of alcohol in road incidents involving both drivers and non-motorists. The overwhelming majority of cases fall under the **“**None Detected**”** category, shown in grey, indicating that in most recorded incidents, no alcohol or substance abuse was identified. This suggests that **substance abuse is not a dominant factor** in the majority of road safety incidents within the dataset.
- However, there are still notable proportions of cases marked as **“Unknown”** (yellow), which points to **data gaps or limitations in substance testing or reporting** at the time of the incident. A smaller slice represents **“Alcohol Detected”** (blue), and an even smaller segment shows **“Alcohol Present”** (orange), indicating that while substance use is a factor in a **minority of cases**, it remains a **serious concern due to its potential impact on crash severity and risk**.
- the chart underscores that while most drivers and non-motorists are not under the influence during incidents, **continued monitoring and preventive strategies targeting substance-related behaviours are still essential**, especially in the context of high-risk behaviours and night-time or high-speed driving.

i. General Description

This chart explores the presence of substance abuse—specifically alcohol—among drivers and non-motorists involved in road incidents. The data categorizes incidents based on substance detection: *Alcohol Detected*, *Alcohol Present*, *None Detected*, and *Unknown*. The goal is to understand how frequently alcohol is a factor in accidents and assess potential gaps in substance-related data collection.

ii. Specific Requirement

The specific objective is to:

- Identify how often alcohol is involved in crashes.
- Distinguish between confirmed alcohol detection, presence, or absence.
- Evaluate the extent of unknown or missing data.
- Use the findings to recommend safety improvements or policy interventions.

iii. Analysis Result

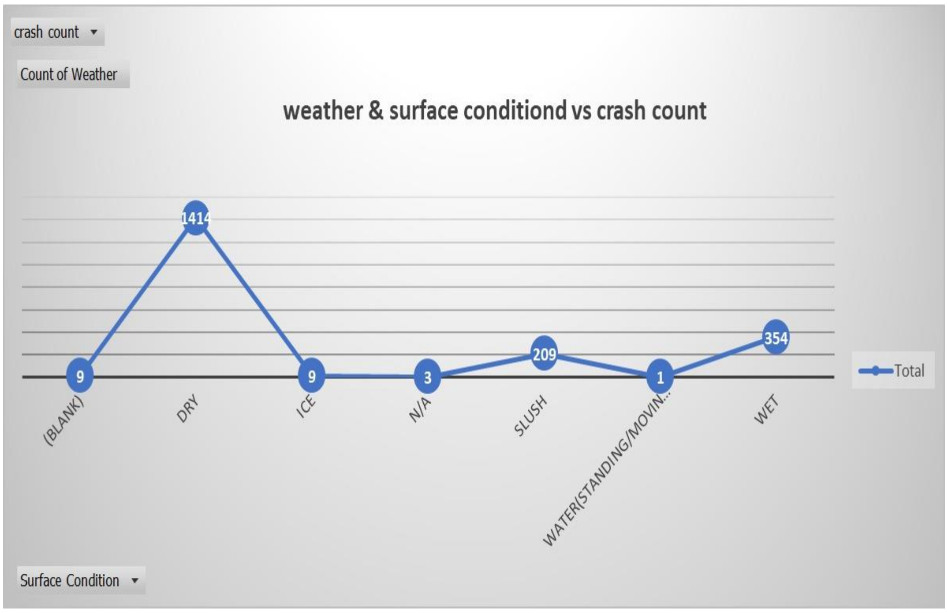
From the chart, it's clear that the majority of incidents show no alcohol involvement, as indicated by the dominant grey segment (*None Detected*). A **significant portion of cases are labelled "Unknown"**, which suggests incomplete reporting or lack of testing. Only a **small fraction of incidents involve confirmed alcohol use** (*Alcohol Detected* and *Alcohol Present*), but even this small percentage is critical, as alcohol-related crashes often result in higher severity outcomes. The data shows that while **substance abuse isn't prevalent**, it remains a **key risk factor** that requires continued monitoring and targeted interventions.

iv. Visualization

The **pie chart** effectively communicates the proportions of each substance abuse category.

- **Grey** represents "None Detected" (majority)
- **Yellow** shows "Unknown" cases (moderate)
- **Blue** indicates "Alcohol Detected" (small)
- **Orange** shows "Alcohol Present" (very small)

2. Weather & surface conditions vs. crash count



- The chart titled **“Weather & Surface Condition vs Crash Count”** provides insightful data on how different road surface conditions correlate with accident frequency. Interestingly, the highest number of crashes—**1,414 incidents**—occurred on **dry roads**, indicating that normal weather conditions do not necessarily guarantee safety. This suggests that **human factors such as speeding, distraction, or traffic volume** might contribute more to accidents than weather alone. The **second highest crash count** is on **wet roads**, with **354 incidents**, highlighting the risks associated with reduced traction and longer braking distances during rainy conditions.
- **Slush-\covered roads** follow with **209 crashes**, pointing to the hazards of snowy or partially melted conditions. Surprisingly, **icy roads only accounted for 9 crashes**, possibly because drivers exercise extra caution or avoid travel during visibly dangerous weather. Similarly, **standing or moving water on roads** led to just **1 crash**, reinforcing the idea that extreme conditions might deter road use entirely. Minor entries like **(BLANK)** and **N/A**, with **9 and 3 crashes** respectively, represent gaps in the dataset. Overall, the data emphasizes that **driver behaviours plays a critical role in road safety**, often more so than environmental factors.
- Based on the line chart titled "weather & surface condition vs crash count", here's a structured response including a general description, specific requirement, analysis result, and visualization insights:

i. General Description

- This chart presents a comparison between different weather and surface conditions and the crash count recorded under each condition. It appears to summarize accident data categorized by weather-related road surface types such as "Dry," "Wet," "Ice," etc. The data points represent how many crashes occurred in each condition.

ii. Specific Requirement

- The objective of this analysis is to:
- Identify which surface conditions are most associated with traffic crashes.
- Help in formulating safety policies, resource allocation, and road maintenance strategies based on environmental factors.

- Support risk assessment models for driving safety under varied weather/surface conditions.

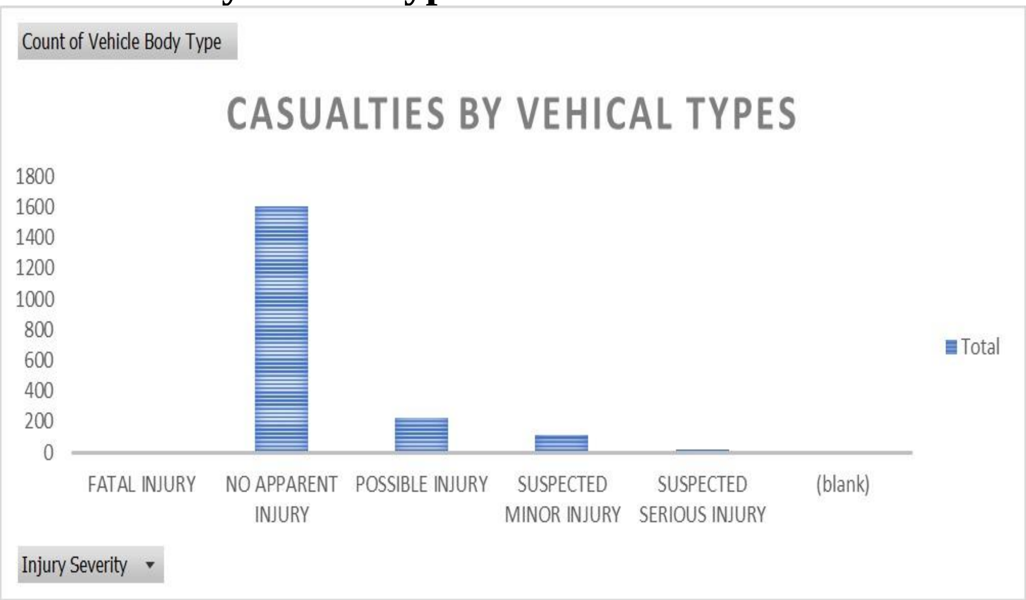
iii. Analysis Result

- Dry surface conditions have the highest number of crashes (1414), which could be due to higher traffic volume under normal conditions.
- Wet surface conditions are the second highest in crash count (354), highlighting potential risk due to reduced friction or visibility.
- Slush (209 crashes) and Ice (9 crashes) are notable for adverse conditions but show lower crash numbers—possibly due to reduced traffic volume during such conditions or effective road closures.
- Minimal crashes are reported under Standing Water, Blank, and N/A, which might reflect either lower exposure or data inconsistencies.

iv. Visualization

- The chart is a line graph, with the x-axis representing various surface conditions and the y-axis showing crash count.
- Data points are labelled for clarity, and there is a significant spike at "Dry" conditions.
- A clear downward trend follows after the peak, with minor rises (e.g., "Slush," "Wet").
- Categories like "Blank," "N/A," and "Water Standing/Moving" show minimal incidents, possibly indicating rare conditions or incomplete reporting,

3. Casualties by vehicle type



- The bar chart titled **"Casualties by Vehicle [Vehicle] Types"** illustrates the distribution of injury severity across various incidents involving vehicles. A striking observation from the chart is that the majority of individuals—**over 1,500 cases**—

sustained **“No Apparent Injury”**, indicating that most accidents resulted in minimal or no physical harm. This could be due to improved vehicle safety features, lower-speed collisions, or the effectiveness of protective gear like seatbelts and airbags.

- A significantly smaller portion of casualties falls under **“Possible Injury”**, followed by **“Suspected Minor Injury”**, both showing moderate figures compared to the dominant "no injury" category. These indicate some level of impact but generally non-life-threatening conditions. Meanwhile, **“Suspected Serious Injury”** and **“Fatal Injury”** account for only a **very small fraction** of the total cases, suggesting that while severe and fatal incidents do occur, they are relatively rare within this dataset. The presence of a **blank category** may point to missing or unreported data, which should be addressed for a more accurate assessment.
- the data implies that **most vehicle-related incidents are not severe**, but the low numbers in serious and fatal categories still highlight the importance of continuous road safety improvements and preventive measures.

i. General Description

- The chart titled "CASUALTIES BY VEHICAL TYPES" shows the distribution of injuries based on vehicle body types. The injuries are categorized by severity levels such as:
- Fatal Injury
- No Apparent Injury
- Possible Injury
- Suspected Minor Injury
- Suspected Serious Injury
- The count on the Y-axis represents the number of casualties across these injury severity levels.

ii. Specific Requirement

- The specific requirement appears to be:
- Understanding the distribution of injury severity due to vehicle incidents.
- Identifying the most common injury types in the dataset.
- Analyzing the safety outcomes of vehicle incidents based on the count of each injury type.

iii. Analysis Result

- "No Apparent Injury" dominates with the highest number of casualties, nearing 1600.

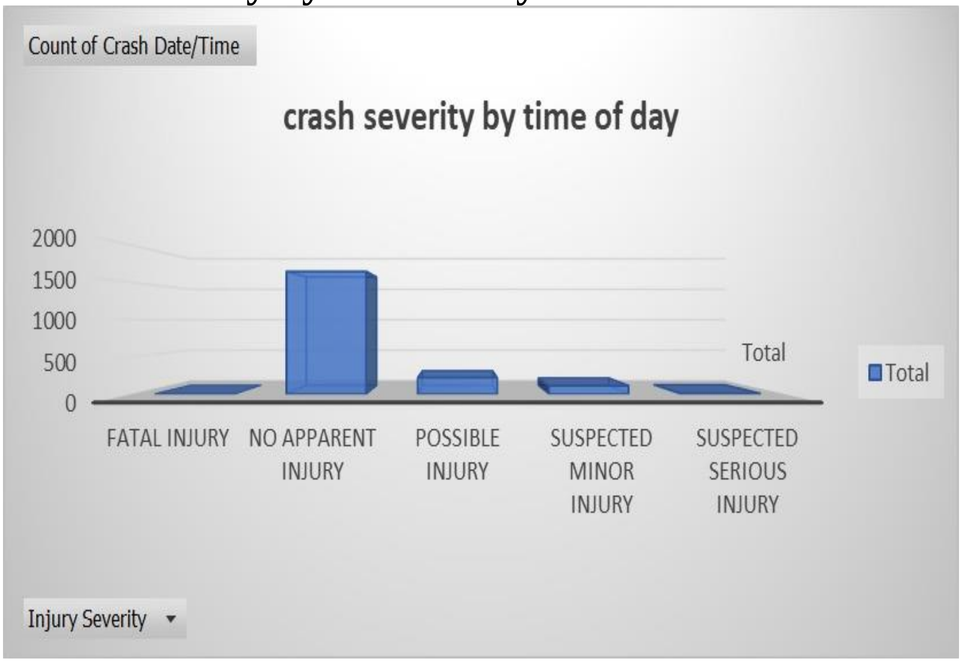
- "Possible Injury" and "Suspected Minor Injury" follow but with significantly lower counts.
- "Fatal Injury" and "Suspected Serious Injury" have very low frequencies.
- A blank category is present, which might indicate missing or unclassified data.

- This suggests that most vehicle-related incidents resulted in no visible injuries, which could imply effective safety measures, minor incidents, or underreporting of actual injuries.

iv. Visualization

- Clustered Column Chart (also known as a Vertical Bar Chart)
- To compare values across categories (in this case, injury severity types).
- Ideal for visualizing discrete categories and their frequencies.

4. Crash severity by time of day



- The bar chart titled **"Crash Severity by Time of Day"** shows the distribution of crash outcomes based on injury severity. The most prominent category is **"No Apparent Injury,"** with a total exceeding **1,500 cases**, indicating that the majority of crashes, regardless of the time they occurred, did not result in visible or reported physical harm. This trend could reflect low-speed collisions, increased use of safety technologies, or prompt medical response minimizing injury impact.

- Following this, **“Possible Injury”** and **“Suspected Minor Injury”** categories show **moderate counts**, suggesting that a smaller yet noteworthy proportion of crashes involved minor or uncertain levels of injury. **“Suspected Serious Injury”** and **“Fatal Injury”** are the least represented, showing that **severe and fatal accidents are relatively rare** compared to less serious outcomes.
- the data indicates that **most crashes are not life-threatening**, and peak crash times may not always correlate with the most severe outcomes. However, even a small number of serious and fatal injuries reinforces the importance of proactive safety strategies, particularly during high-risk hours like night or early morning when driver alertness might be lower.

i. General Description

- The visualization presents a summary of crash incidents categorized by the severity of injury and based on the time of day. The data focuses on five injury severity categories: Fatal Injury, No Apparent Injury, Possible Injury, Suspected Minor Injury, and Suspected Serious Injury

ii. Specific Requirement

- Understand the distribution of traffic crash severity over time.
- Identify which types of injuries are most and least frequent.

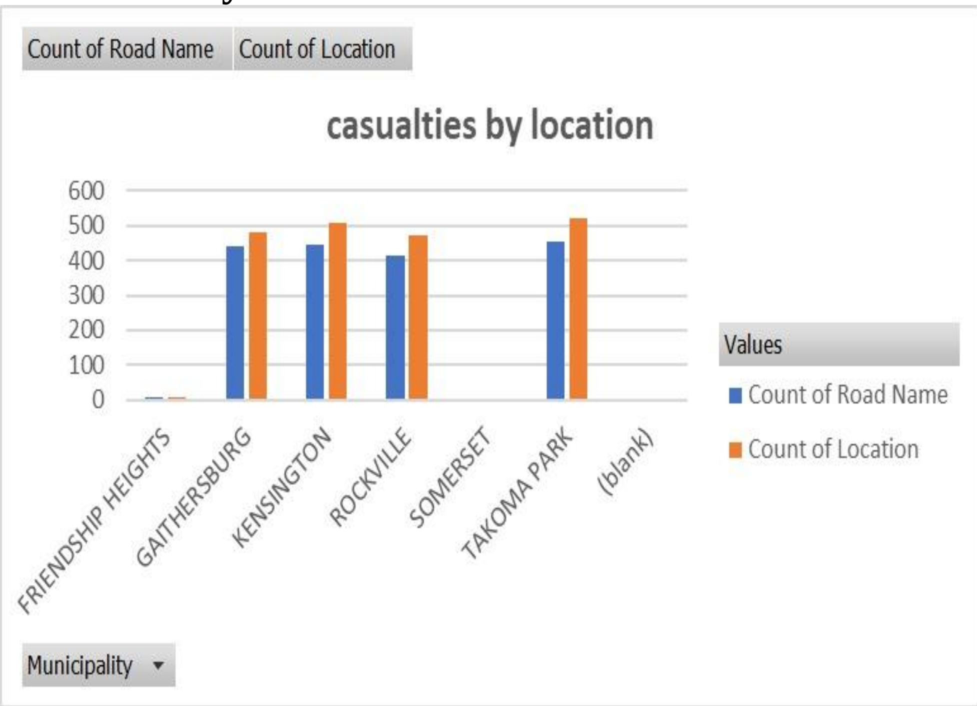
ii. Analysis Result

- No Apparent Injury is the most common outcome, with a count close to 1800.
- Possible Injury and Suspected Minor Injury follow next, each with significantly fewer counts (under 300).
- Suspected Serious Injury and Fatal Injury are the least common, both appearing in very small numbers.

iii. Visualization

- X-Axis: Injury severity categories.
- Y-Axis: Count of crash incidents.
- Bars: Represent the total number of crashes per severity type. The height of the bars clearly shows that "No Apparent Injury" dominates the dataset.

5. Casualties by location



- The chart titled **"Casualties by Location"** compares casualty counts across different municipalities using two metrics: **Count of Road Name** and **Count of Location**. The municipalities of **Gaithersburg, Kensington, Rockville, Somerset, and Takoma Park** all show relatively **high and consistent casualty figures**, generally ranging between **400 to over 500 incidents**. This indicates that these areas experience a **significant share of road incidents**, which may be due to factors such as traffic density, urban infrastructure, or population concentration.
- Among the locations, **Kensington and Takoma Park** stand out with the **highest counts**, suggesting they may be key areas of concern for road safety interventions. Interestingly, **Friendship Heights** reports **negligible casualties**, which could imply either fewer road incidents or underreporting. The chart also includes a **"(blank)" category**, representing data entries with missing or unassigned locations, which surprisingly shows **high casualty counts as well**. This highlights a gap in data quality and underlines the importance of **accurate geographic tagging** for effective analysis and targeted safety measures.
- The chart emphasizes that **location plays a crucial role** in the frequency of casualties, and focused efforts in high-incidence areas could significantly improve road safety outcomes.

I. General Description

This dataset represents casualty counts in various municipalities. Specifically, it includes a pivot table showing the number of casualties by Road Name and Location for each area. The table also includes a chart summarizing this information visually.

ii. Specific Requirement

- Objective: To analyse and compare the number of casualties in different municipalities.
- Row Labels: Municipality names
- Values:
- Count of Road Name (possibly represents the number of roads involved in incidents)
- Count of Location (possibly represents the number of incident spots or events)
- Goal: Identify which municipalities have the highest and lowest casualty-related counts.

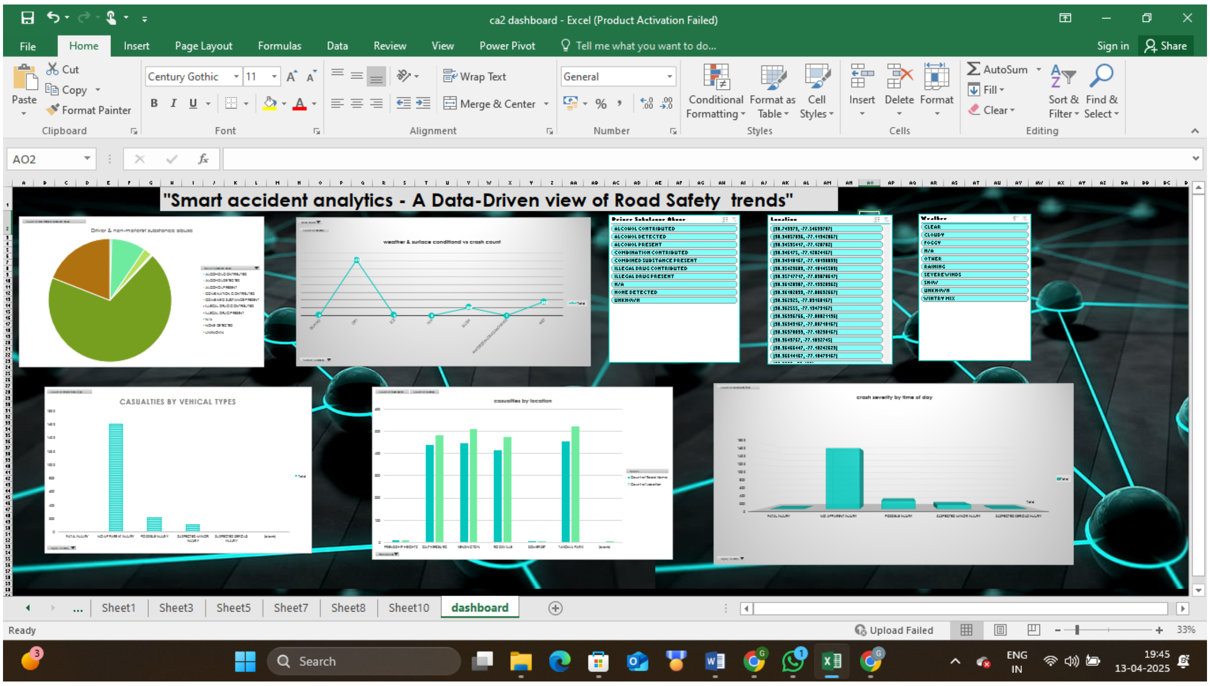
iv. Analysis Result

- From the pivot table:
- Highest Casualties:
- Takoma Park: 453 (Roads), 522 (Locations)
- Kensington: 446 (Roads), 510 (Locations)
- Gaithersburg: 439 (Roads), 483 (Location)

v. Visualization

- Chart Type Used: Clustered Column Chart
- Purpose: To compare casualty counts (by road name and location) across municipality's side-by-side.
- Vertical bars for each municipality
- Two colored bars per municipality to distinguish between "Count of Road Name" and "Count of Location"

Dashboard


ca2 dashboard - Excel (Product Activation Failed)
"Smart accident analytics - A Data-Driven view of Road Safety trends"
CASUALTIES BY VEHICAL TYPES
casualties by location
crash severity by time of day
weather & surface conditiond vs crash count
Sheet1
Sheet3
Sheet5
Sheet7
Sheet8
Sheet10
dashboard

Conclusion

- Smart Accident Analytics represents a significant advancement in the way road safety is understood and managed. By leveraging vast amounts of data and cutting-edge analytical tools, this approach provides valuable insights into accident trends, risk factors, and driver behaviours. It allows for a shift from reactive responses to proactive, evidence-based interventions aimed at reducing accidents and saving lives.
- As urban environments continue to grow more complex, the use of smart analytics in transportation becomes increasingly essential. While there are challenges—such as ensuring data privacy, maintaining system accuracy, and integrating diverse data sources—the benefits are substantial. Enhanced traffic management, faster emergency response, and informed infrastructure planning are just a few of the outcomes made possible through data-driven insights.
- In conclusion, Smart Accident Analytics is not just a technological innovation, but a necessary evolution in road safety strategy. Its implementation paves the way for safer, more efficient, and smarter transportation systems, ultimately contributing to a reduction in accidents and improved public safety on the roads.